

# Virtual Shopping Complex

AR/VR College Project · Unity 6 LTS

Download & Setup Guide



*The shopping complex exterior (preview render of the project layout).*

| What you get |                                                                                                                                                                 |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project      | A complete Unity project: a first-person shopping mall with 6 shops and 132 interactive products, made only from Unity primitives (no paid or external assets). |
| Controls     | WASD move · Mouse look · Space jump · Shift run · E interact · Esc unlock cursor                                                                                |
| Time needed  | About 30–60 minutes the first time (most of it is installing Unity).                                                                                            |

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## 1. What you need

| Item          | Details                                                                                                                     |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| Computer      | Windows 10/11 (64-bit) or macOS 12+. Any normal laptop/desktop with 8 GB RAM or more.                                       |
| Disk space    | About 10 GB free (Unity editor ~6-8 GB, project ~1 GB after import).                                                        |
| Internet      | Needed once to download Unity Hub, the Unity editor and the project.                                                        |
| Unity account | Free Unity ID (sign up inside Unity Hub). The free <b>Personal</b> licence is enough.                                       |
| Unity version | <b>Unity 6 LTS</b> (version number starts with 6000.0 or 6000.3). Any Unity 6 LTS works.                                    |
| Packages      | None to install manually. The project's Packages/manifest.json already lists everything (Unity UI + optional Input System). |

## 2. Download the project

Pick **one** of the three options. All of them give you a folder called **VirtualShoppingComplex** that contains the folders **Assets**, **Packages** and **ProjectSettings**. That folder is your Unity project.

### Option A - Download the ZIP from the Claude chat (easiest)

1. In the Claude conversation, find the file card **VirtualShoppingComplex.zip** and click it to download (on a phone, open the card and use the download/share button).
2. Also download **Virtual\_Shopping\_Complex\_Setup\_Guide.pdf** (this file) and **virtual\_shopping\_complex\_walkthrough.mp4** (the preview video) if you want them.
3. Move the ZIP to an easy place, for example **Documents\UnityProjects**.
4. Right-click the ZIP → **Extract All...** (Windows) or double-click it (Mac).
5. Open the extracted folder and check that you see **VirtualShoppingComplex** with **Assets**, **Packages** and **ProjectSettings** inside.

## Option B - Download from GitHub in a browser

1. Sign in to GitHub and open **<https://github.com/patrickmathew0640-creator/cloud>**.
2. Click the branch selector (it says **main** or similar) and choose **claude/shopping-complex-unity-p637qt**.
3. Click the green **Code** button → **Download ZIP**.
4. Extract the ZIP. Inside you will find the **VirtualShoppingComplex** folder (the repository also contains other folders – you can ignore them).
5. The video is stored with Git LFS, so GitHub's ZIP only contains a small placeholder for it. To get the real video, open **deliverables/virtual\_shopping\_complex\_walkthrough.mp4** on GitHub and click **Download raw file**.

## Option C - Clone with Git (for developers)

```
git lfs install
git clone --branch claude/shopping-complex-unity-p637qt \
  https://github.com/patrickmathew0640-creator/cloud.git
cd cloud/VirtualShoppingComplex
```

**Tip:** keep the project path short and without special characters (e.g. **C:\UnityProjects\VirtualShoppingComplex**). Very long Windows paths and cloud-synced folders (OneDrive) can cause import problems.

## 3. Install Unity Hub and Unity 6 LTS

1. Go to **[unity.com/download](https://unity.com/download)** and install **Unity Hub**.
2. Open Unity Hub and sign in (create a free Unity ID if needed). Accept the free **Personal** licence when asked.
3. In Unity Hub go to **Installs** → **Install Editor**.
4. Choose the newest **Unity 6 ... LTS** version under *Official releases*.
5. In the modules list tick **Microsoft Visual Studio Community** (optional, for editing scripts) and **Windows Build Support (Mono)** (only needed if you want to build an .exe). Click **Install**.
6. Wait for the installation to finish (10–30 minutes depending on your internet).

## 4. Open the project

1. In Unity Hub click **Projects** → **Add** (or the arrow next to it) → **Add project from disk**.
2. Select the **VirtualShoppingComplex** folder (the one that directly contains Assets, Packages, ProjectSettings) and click **Add Project / Open**.
3. If the Hub shows a warning that editor version **6000.0.23f1** is not installed, click the version number and pick the Unity 6 LTS version you installed, then confirm **Change version / Continue**.
4. Click the project to open it. The first import takes **3-10 minutes**. Do not close Unity while it imports.

5. If a dialog asks “...native platform backends for the new input system are not enabled... Do you want to enable the backends?” click **Yes**. Unity restarts once. (Clicking **No** also works – the scripts support both input systems.)
6. When the editor opens, the scene **Assets/Scenes/VirtualShoppingComplex.unity** is created and opened automatically. The Hierarchy window shows one object called **VirtualShoppingComplex**.
7. If the scene is not open, use the top menu **Virtual Mall → Open or Create Mall Scene**.

**Do not worry that the Scene view looks empty.** The whole mall, the player and the UI are generated automatically when you press Play. This avoids broken references and missing scripts. If you want to see and edit everything in Edit Mode, see section 8 (Bake Mall Into Scene).

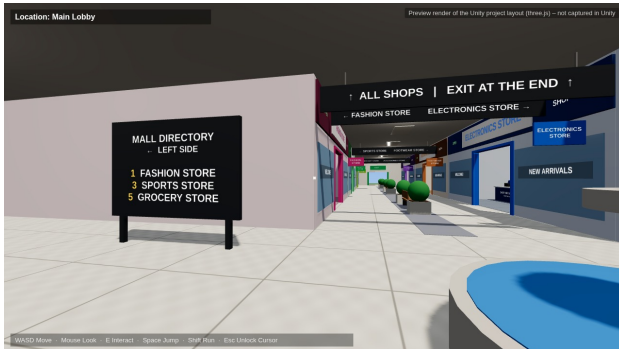
## 5. Play the demo

1. (Recommended) In the **Game** tab, set the aspect drop-down to **16:9** or **Full HD (1920×1080)** and enable **Maximize On Play** (or **Play Maximized**).
2. Press the ► **Play** button at the top of the editor.
3. The start menu appears. Click **START TOUR** or press **Enter**.
4. Click once inside the Game view so the mouse cursor is captured, then explore.
5. Press ► **Play** again to stop.

| Key                  | Action                                                                                          |
|----------------------|-------------------------------------------------------------------------------------------------|
| W A S D / Arrow keys | Move                                                                                            |
| Mouse                | Look around                                                                                     |
| Space                | Jump                                                                                            |
| Shift (hold)         | Run                                                                                             |
| E                    | Interact with the product or shop counter you are looking at; also closes the information panel |
| Esc                  | Unlock the mouse cursor, or close the open panel                                                |
| Left mouse click     | Lock the cursor again                                                                           |

## 6. Demo walkthrough (what you should see)

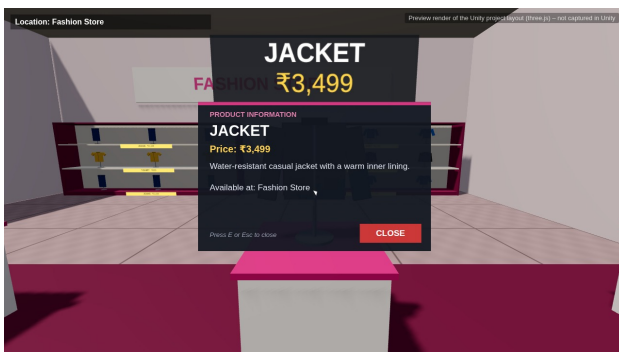
The images in this guide come from a **preview render** of the project's layout (made with three.js, because the cloud machine that prepared this guide has no Unity installed). Your Unity screen should look very similar – lighting and fonts can differ slightly.



Lobby: directory boards, fountain and the start of the central corridor.



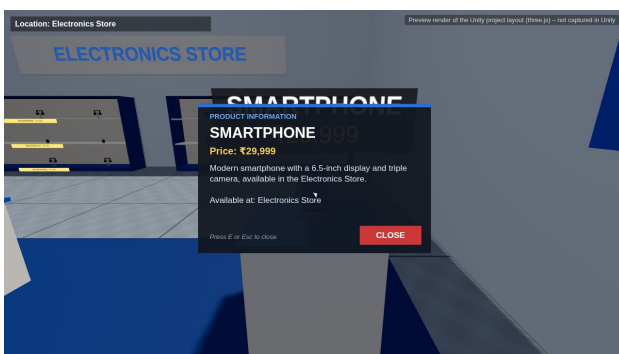
Fashion Store: podiums with name/price labels and stocked shelves.



Look at a product and press E: name, price and description appear.



Look at a shop counter: "Press E to Explore Shop".



Electronics Store: Smartphone – ₹29,999.

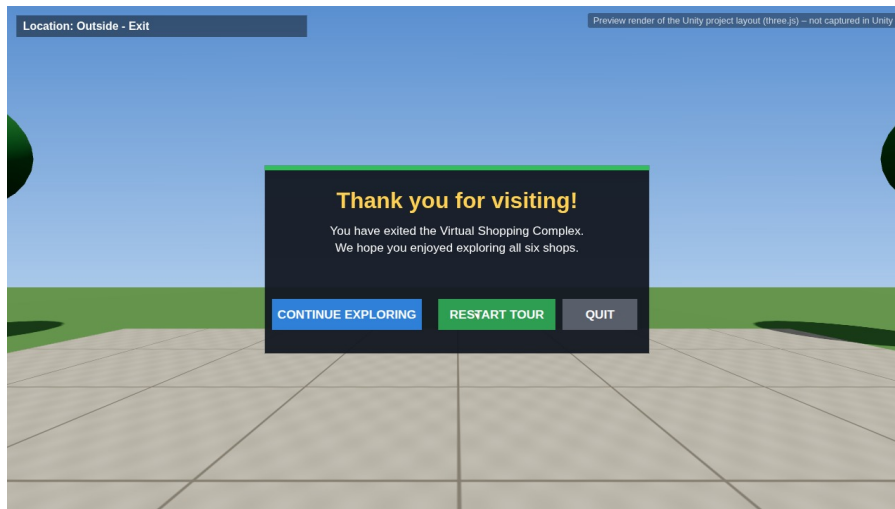


Grocery Store shelves with price tags on every shelf.

### Complete demo flow

1. Start menu → START TOUR → "Welcome to the Virtual Shopping Complex" (disappears after ~4 s).
2. You stand outside the **main entrance**. Walk in → **Main Lobby** → **Central Corridor**.
3. Left side: 1 Fashion, 3 Sports, 5 Grocery. Right side: 2 Electronics, 4 Footwear, 6 Accessories.
4. Inside a shop, look at a product → **Press E to Interact** → product panel → close with **CLOSE, E or Esc**.

5. Look at the counter → **Press E to Explore Shop** → shop name, number of products, description.
6. At the end of the corridor is the **Exit Hall**. Walk out through the green **EXIT** door → “Thank you for visiting!” → Continue Exploring / Restart Tour / Quit.



*Exit screen after leaving through the back EXIT door.*

## Shops and products

| Shop        | Products (price)                                                                 |
|-------------|----------------------------------------------------------------------------------|
| Fashion     | Shirt ₹1,299 · Jacket ₹3,499 · Jeans ₹2,199 · T-Shirt ₹599                       |
| Electronics | Smartphone ₹29,999 · Laptop ₹59,999 · Headphones ₹4,999 · Smartwatch ₹8,999      |
| Sports      | Football ₹1,499 · Basketball ₹1,799 · Sports Shoes ₹3,999 · Tennis Racket ₹2,499 |
| Footwear    | Sneakers ₹2,999 · Running Shoes ₹4,499 · Sandals ₹899 · Boots ₹5,499             |
| Grocery     | Juice Bottle ₹120 · Cereal Box ₹349 · Milk Bottle ₹65 · Snack Packet ₹40         |
| Accessories | Watch ₹6,999 · Sunglasses ₹1,999 · Backpack ₹2,499 · Wallet ₹999                 |

To change names, prices, descriptions or colours, edit **Assets/Scripts/Environment/MallCatalog.cs** and press Play again.

## 7. Project structure

```
VirtualShoppingComplex/  
  Assets/  
    Scenes/          VirtualShoppingComplex.unity (created automatically)  
    Scripts/  
      Core/          InputReader.cs, MallBootstrap.cs  
      Environment/    MallBuilder.cs, MallCatalog.cs, MallAssets.cs, ProductFactory.cs  
      Player/         PlayerController.cs, MouseLook.cs  
      Interaction/    IInteractable.cs, InteractionSystem.cs, InteractableProduct.cs,  
                     ShopInteraction.cs, MallExit.cs  
      UI/             UIManager.cs  
      Editor/         MallSceneSetup.cs (menu "Virtual Mall")  
      Prefabs/ Materials/ Textures/ (filled by "Bake Mall Into Scene")  
      Models/ UI/      (empty - everything is built from primitives/code)  
  Packages/manifest.json  
  ProjectSettings/ProjectVersion.txt
```

| Script              | Purpose                                                                                        |
|---------------------|------------------------------------------------------------------------------------------------|
| MallBootstrap       | Only component in the scene. On Play it builds the whole mall, player and UI.                  |
| MallBuilder         | Creates plaza, building, entrance, lobby, corridor, 6 shops, exit hall, lights, player and UI. |
| MallCatalog         | All shop/product data (names, prices, descriptions, colours); Indian price format.             |
| MallAssets          | Primitive meshes, materials (Built-in & URP), font, floor texture, world-space labels.         |
| ProductFactory      | Builds the 24 primitive product models.                                                        |
| InputReader         | Reads keyboard/mouse for both the old Input Manager and the new Input System.                  |
| PlayerController    | CharacterController movement: walk, run, jump, gravity, collisions, respawn.                   |
| MouseLook           | First-person mouse look with smoothing and clamped up/down angle ( $\pm 85^\circ$ ).           |
| InteractionSystem   | Raycast from the camera; shows the prompt and calls Interact() on E.                           |
| InteractableProduct | Product data and the product information panel.                                                |
| ShopInteraction     | Shop counter: shop information panel and the shop area for the Location HUD.                   |
| MallExit            | Exit zone outside the back door that opens the "Thank you" screen.                             |
| UIManager           | Builds and controls all UI: start menu, welcome, prompt, panels, exit screen, cursor lock.     |
| MallSceneSetup      | Editor menu: create/open scene, add to Build Settings, Bake/Clear mall.                        |

## 8. Optional extras

### See and edit the mall in Edit Mode + create real prefabs

1. Open the mall scene (not in Play mode).
2. Menu **Virtual Mall → Bake Mall Into Open Scene (editable objects + prefabs)**.
3. The whole hierarchy (Environment, Shops, Player, UI) now appears in the Hierarchy. Materials are saved to **Assets/Materials**, the floor texture to **Assets/Textures**, and



prefabs to **Assets/Prefabs** (24 products, 6 shops, Player).

4. To go back to the automatically generated version: **Virtual Mall → Clear Baked Mall From Open Scene**.

## Build a Windows .exe for your presentation

1. **File → Build Profiles** (older versions: **Build Settings**).
2. Select **Windows** and check that **Scenes/VirtualShoppingComplex** is in the scene list (it is added automatically; otherwise click **Add Open Scenes**).
3. Click **Build**, choose an empty folder, wait, then run **VirtualShoppingComplex.exe**.

## Record your own video from Unity

- **Windows:** press **Win + Alt + R** (Xbox Game Bar) while the Game view is focused; press again to stop.
- **Mac:** press **Cmd + Shift + 5** → Record Selected Portion.
- **Inside Unity:** Window → Package Manager → Unity Registry → install **Recorder**, then Window → General → Recorder → Recorder Window → add **Movie** → press **Start Recording**.

## 9. Converting to VR (OpenXR)

The desktop version works without any XR packages. When you are ready for VR:

1. Window → Package Manager → Unity Registry: install **XR Plugin Management**, **OpenXR Plugin** and **XR Interaction Toolkit** (then import its **Starter Assets** sample).
2. Edit → Project Settings → XR Plug-in Management: tick **OpenXR**; add your headset's interaction profile (e.g. Oculus Touch Controller Profile).
3. Bake the mall (section 8), then disable the baked **Player** object.
4. Drag the **XR Origin (XR Rig)** prefab into the scene at position (0, 0, -7).
5. On the controller's ray interactor, on *Select Entered* call **Interact()** on the **IInteractable** of the selected object (same method the E key uses).
6. Switch the UIManager canvas to **World Space**, place it ~2 m in front of the player and add a **Tracked Device Graphic Raycaster**.

## 10. Final test checklist

| ✓                        | Check                                                                                                                |
|--------------------------|----------------------------------------------------------------------------------------------------------------------|
| <input type="checkbox"/> | Project opens with no red errors in the Console (Window → General → Console).                                        |
| <input type="checkbox"/> | Scene VirtualShoppingComplex opens; Hierarchy shows VirtualShoppingComplex with Mall Bootstrap, no "Missing script". |
| <input type="checkbox"/> | Play → start menu with VIRTUAL SHOPPING COMPLEX and the CONTROLS list.                                               |
| <input type="checkbox"/> | START TOUR → welcome message shows and fades out after ~4 seconds.                                                   |
| <input type="checkbox"/> | Player starts outside the main entrance.                                                                             |
| <input type="checkbox"/> | WASD moves, Shift runs, Space jumps.                                                                                 |
| <input type="checkbox"/> | Mouse look works; looking fully up/down stops at the limit and never flips.                                          |



| ✓                        | Check                                                                                         |
|--------------------------|-----------------------------------------------------------------------------------------------|
| <input type="checkbox"/> | Cannot walk through walls, counters, shelves, podiums, benches, fountain or boards.           |
| <input type="checkbox"/> | Location display (top-left) updates: Main Lobby, Central Corridor, shop names, Exit Hall.     |
| <input type="checkbox"/> | All six shop names visible; every product has a name and price label.                         |
| <input type="checkbox"/> | “Press E to Interact” appears on products; E opens name/price/description.                    |
| <input type="checkbox"/> | Panel closes with CLOSE, E or Esc and does not reopen by itself.                              |
| <input type="checkbox"/> | “Press E to Explore Shop” on counters; E shows shop name, number of products, description.    |
| <input type="checkbox"/> | Esc unlocks the cursor (orange banner); left click locks it again.                            |
| <input type="checkbox"/> | Walking out of the EXIT door shows “Thank you for visiting!”; Continue / Restart / Quit work. |

## 11. Troubleshooting

| Problem                                                   | Solution                                                                                                                                 |
|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Unity Hub cannot find the project / “not a valid project” | Select the <b>VirtualShoppingComplex</b> folder itself (the one containing Assets, Packages, ProjectSettings), not the outer ZIP folder. |
| Editor version not installed                              | Click the version in Hub and pick your installed Unity 6 LTS.                                                                            |
| Mouse does not turn the camera                            | Close the start menu (START TOUR) and click inside the Game view.                                                                        |
| Mouse too fast/slow                                       | During Play select Player → PlayerCamera and change Mouse Look → Sensitivity (default 2).                                                |
| Input System dialog on first open                         | Click Yes (recommended) or No – both work.                                                                                               |
| Pink / magenta objects                                    | Only after baking with a changed render pipeline: run Clear Baked Mall, then Bake again (or just use Play without baking).               |
| Scene not opened automatically                            | Menu Virtual Mall → Open or Create Mall Scene.                                                                                           |
| “Virtual Mall” menu missing                               | Wait for script compilation to finish (spinner bottom-right); check the Console for red errors.                                          |
| Prices show “Rs.” instead of ₹                            | Normal: the built-in font on that computer has no ₹ glyph, so the project falls back automatically.                                      |
| Red compile errors                                        | Copy the first red error from the Console and send it – it can be fixed quickly.                                                         |